Metal Prices in Transition: Structural Tightness and the Long-Term Outlook for Copper and Silver

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Historical price evidence, inventory signals, return correlations, and long-run uncertainty bands

RESEARCH QUESTION

Are copper and silver entering a tighter long-run regime than other major metals?

This project separates three forces:
- short-run macro conditions
- physical-market tightening
- structural demand tied to electrification

DATA

- World Bank monthly prices for six metals, 2000–2026
- FRED and OECD macro controls
- LME copper stocks
- IEA, ICSG, Silver Institute, and USGS structural inputs

APPROACH

- Compare indexed price paths across metals
- Measure monthly return correlations
- Test whether copper prices rise as exchange stocks fall
- Extend the paper with scenario paths and a 10,000-draw Monte Carlo

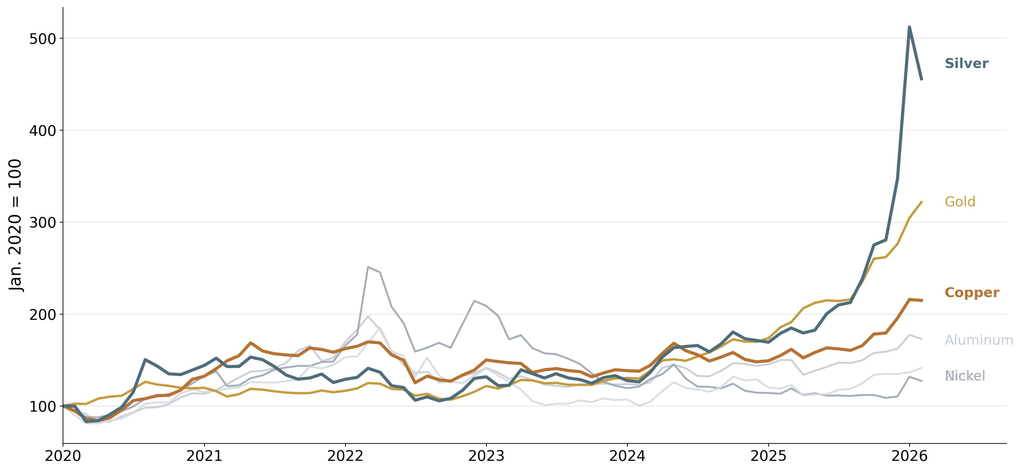
WHAT TO NOTICE

- Copper and silver pull away after 2020
- Copper tightening shows up in inventories, not just price
- Silver remains a hybrid metal
- Long-run ranges widen over time, especially for silver

FIGURE 1

Copper and silver pulled away after 2020

Indexed prices, Jan. 2020 = 100

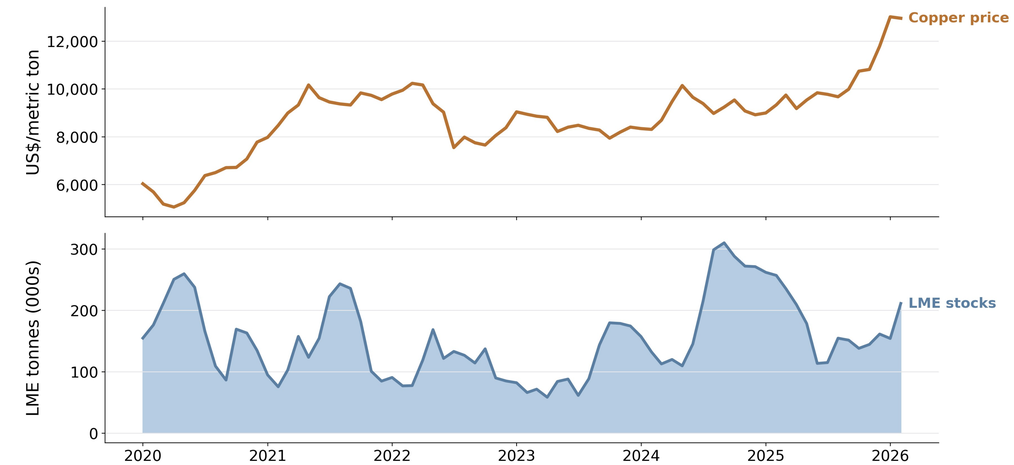


Source: World Bank Pink Sheet monthly prices; author calculations.

FIGURE 2

Copper rose as exchange stocks fell

Monthly data, 2020–2026

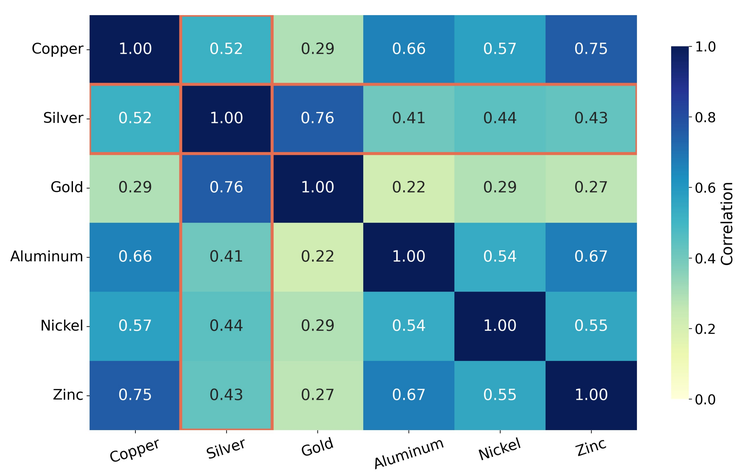


Source: World Bank Pink Sheet; LME/Westmetall; author calculations.

FIGURE 3

Silver sits between precious and industrial metals

Monthly return correlations, 2010–2025

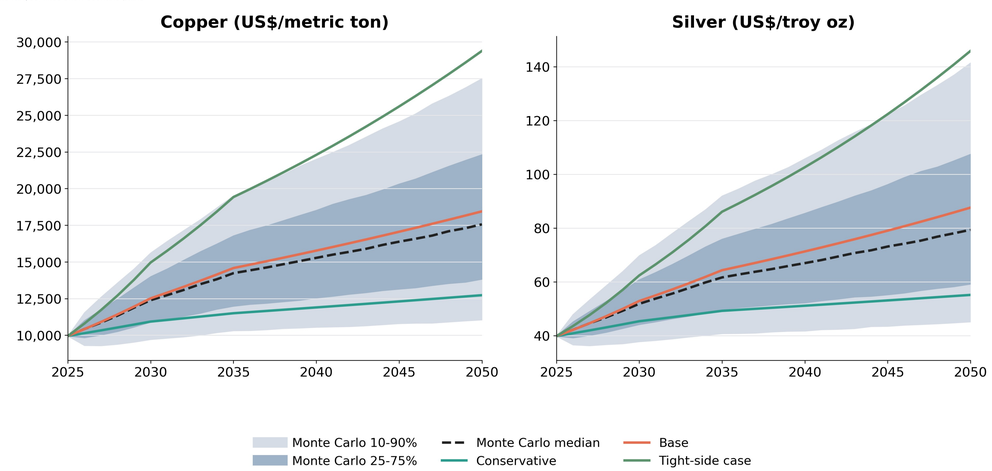


Source: World Bank Pink Sheet monthly returns; author calculations.

FIGURE 4

Scenario paths sit inside widening Monte Carlo ranges

Bands show Monte Carlo ranges; lines show scenario paths



Source: project structural model and 10,000-path Monte Carlo simulation.

Selected sources: World Bank Pink Sheet (2026); FRED/OECD (2026); LME/Westmetall; IEA (2025); ICSG (2025a, 2025b); Silver Institute (2025a–2025c); USGS (2026).